

CSS 324: Introduction to Machine Learning - Final Project

Team Registration

Students may work **individually or in teams of 4–5 students**.

Each team must register by filling out the [Google Form](#) provided by the instructor.

Registration deadline: March 29, 2026 (Week 10)

Important notes:

- Only **one student per team** should submit the form.
- All team members must be listed in the submission.
- Make sure the team **agrees on the project topic before submitting the form**.
- If you do not have a team by the deadline, the instructor will **assign you to a team randomly**.

This form is used only for [team registration and project topic approval](#).

Project Requirements

1. Topic

Select a machine learning problem that solves a real-world task. Clearly explain why the problem is important and how machine learning can help.

Example topics: image classification, text analysis, recommendation systems, prediction, clustering, anomaly detection.

2. Data Collection & Preparation

Use a dataset that requires data cleaning and preprocessing. Raw or partially unclean data is encouraged. Datasets may come from Kaggle, public repositories, APIs, or be collected from multiple sources. Document: dataset source, data cleaning steps, preprocessing and feature engineering

Projects that use raw or non-ready datasets and require significant data cleaning, preprocessing, or combining multiple data sources will receive the highest scores.

3. Research & Data Analysis (EDA)

- a. Review **2–3 related papers or articles**.
- b. Perform exploratory analysis, including: statistical summaries, visualizations, interpret patterns and insights

4. Machine Learning Experiments

Train **at least 3 different models**. Tune hyperparameters and evaluate performance using appropriate metrics. Compare models and justify your **final model selection**. Analyze errors and limitations.

5. Interactive Demonstration Application

Students must demonstrate how their final model can be used to make predictions. This can be implemented using:

- an interactive Jupyter Notebook
- a Python script with user input
- an interactive application using Streamlit or Gradio

All projects must be submitted through a **GitHub repository**

6. Presentation & Report

Presentation (5–7 minutes): logical flow, clarity, timing, speaking.

Printed or PDF report: must include all sections: Topic & motivation, Dataset & preparation, EDA, ML experiments, Final model & demo description, Analysis & conclusions

7. Q/A

Be prepared to answer questions about your project, dataset, preprocessing steps, models, evaluation results, and the demonstration application.